CYBERSECURITY LAB REPORT

RDP Brute-Force Attack — Detection & Incident Response

Portfolio Project | March 18, 2026

Tools Used: Kali Linux • Hydra • Splunk SIEM • Windows Event Viewer

Environment: Splunk Universal Forwarder • Windows 10 Target • VirtualBox Lab

EXECUTIVE SUMMARY

This report documents a complete end-to-end cybersecurity lab exercise simulating an RDP (Remote Desktop Protocol) brute-force attack and its detection using SIEM tooling. The attacker machine (Kali Linux) launched a dictionary-based credential attack against a Windows target using Hydra. Detection and analysis were performed using Splunk Enterprise with Windows Security event logs forwarded via the Splunk Universal Forwarder. Key Windows Event IDs (4624, 4625) were used to trace the attack lifecycle, from failed login attempts through to successful authentication. This project demonstrates practical blue-team skills including log analysis, threat detection, and incident response.

1. Project Overview

1.1 Objectives

This lab project was designed to achieve the following learning outcomes in a controlled, ethical lab environment:

- Simulate a real-world RDP brute-force attack using Hydra on Kali Linux
- Configure log forwarding from a Windows host to Splunk via the Universal Forwarder
- Detect and investigate the attack using Windows Security Event IDs in Splunk
- Correlate multiple event types to reconstruct the attack timeline
- Produce documentation suitable for a cybersecurity portfolio

1.2 Lab Environment

Component	OS / Version	Role	IP Address
Attacker Machine	Kali Linux (VirtualBox)	Launches Hydra RDP brute-force	10.0.2.x
Target Machine	Windows 10 (LAPTOP-SMMVETKR)	RDP target, log source	10.0.2.15
Secondary Host	Windows 10 (DESKTOP-HMUS15L)	Additional monitored host	Local
SIEM	Splunk Enterprise	Log aggregation & analysis	localhost:8000
Log Forwarder	Splunk Universal Forwarder	Ships Windows Security logs	Installed on target

1.3 Attack & Detection Methodology

Phase	Name	Description
1	Reconnaissance	Identified target IP (10.0.2.15) with RDP port 3389 open
2	Brute Force Attack	Used Hydra with rockyou.txt wordlist to attempt RDP login as 'admin'
3	Log Collection	Splunk Universal Forwarder shipping WinEventLog:Security to SIEM
4	Detection	Searched Event Code 4625 (failed logon) to identify attack pattern

Phase	Name	Description
5	Confirmation	Event Code 4624 (successful logon) confirmed attacker gained access
6	Post-Incident Review	Windows Event Viewer cross-validated SIEM findings

2. Attack Execution — Hydra RDP Brute-Force

2.1 Tool Used: Hydra

Hydra (THC-Hydra) is an open-source parallelized network login cracker supporting dozens of protocols. In this lab, it was used to perform a dictionary-based brute-force attack against the Windows RDP service on port 3389.

2.2 Attack Command

```
hydra -l admin -P /usr/share/wordlists/rockyou.txt rdp://10.0.2.15
```

2.3 Command Breakdown

Parameter	Explanation
-l admin	Specifies a single username: 'admin' — the account being targeted
-P /usr/share/wordlists/rockyou.txt	Password list: rockyou.txt contains 14+ million commonly used passwords
rdp://10.0.2.15	Protocol and target: RDP protocol against Windows host at 10.0.2.15

2.4 Attack Output Observations

The Kali terminal output showed Hydra v9.6 running at 02:02:39 on 2026-03-18 with the following notable behaviours:

- Tasks automatically reduced to 4 parallel threads (RDP limitation)
- Total login attempts estimated at 14,344,399 (full rockyou.txt)
- Repeated messages: account might be valid but not active for remote desktop
- Intermittent FreeRDP connection failures observed — typical in RDP brute-force
- Passwords attempted included: password, 123456, 123456789, iloveyou, princess, rockyou, abc123, nicole, daniel

Attack Note
The Hydra output showing 'account might be valid but not active for remote desktop' indicates the username 'admin' exists on the target system but Network Level Authentication (NLA) or RDP group

policy restrictions were partially limiting direct access — yet logon events were still generated and captured in the Security log.

3. Log Forwarding — Splunk Universal Forwarder

3.1 Configuration

The Splunk Universal Forwarder was installed on the Windows target machine (LAPTOP-SMMVETKR) and configured to forward Windows Security event logs to Splunk Enterprise. The forwarder was verified to be running before the attack commenced.

3.2 Forwarder Verification Command

```
cd "C:\Program Files\SplunkUniversalForwarder\bin"
splunk start
The splunk daemon (splunkd) is already running.
```

3.3 Log Source Configuration

Setting	Value
Source Type	WinEventLog:Security
Index	Default (main)
Host	LAPTOP-SMMVETKR
Monitored Log	Windows Security Event Log
Forwarding Port	9997 (default Splunk receive)

4. SIEM Detection — Splunk Analysis

4.1 Windows Security Event IDs Used

Event ID	Event Name	Security Significance
4624	An account was successfully logged on	Indicates successful authentication — potential attacker access confirmed
4625	An account failed to log on	Key indicator of brute-force — multiple rapid failures from same source
4627	Group membership information	Logged at logon — shows group membership of logged-on user
4672	Special privileges assigned	Elevated token assigned — indicates admin-level logon
5379	Credential Manager credentials read	Credential access activity — can indicate post-compromise credential harvesting

4.2 Detecting Failed Logons — Event Code 4625

```
EventCode=4625
```

Search results — Last 60 minutes timeframe:

Metric	Value
Total Events Found	1 event (initial 60-min window)
Extended Search (7 days)	12 events total
Hosts Affected	DESKTOP-HMUS15L
Time of First 4625 (60-min)	3/18/2026 01:16:51 AM
Time of Last 4625 (7-day)	3/18/2026 02:54:32 AM
Source	WinEventLog:Security

Detection Finding
The 12 Event 4625 (failed logon) events across the 7-day window, with a cluster of events at 02:54 AM on 18-03-2026 on host DESKTOP-HMUS15L, strongly indicates brute-force activity. Multiple

failed logon attempts in rapid succession with the same target account are a classic brute-force signature.

4.3 Detecting Successful Logons — Event Code 4624

```
EventCode=4624
```

Metric	Value
Total Events (60-min window)	121 events
Total Events (5-min focused)	2 events (around attack time)
Primary Host	LAPTOP-SMMVETKR
Most Recent 4624	3/18/2026 02:12:38 AM
Previous Logon	3/18/2026 02:10:33 AM
Account Names Observed	11 unique accounts
Account Domains Observed	7 unique domains

Critical Finding
Event Code 4624 (successful logon) occurring shortly after the brute-force attack window indicates the attacker successfully authenticated. The 121 events in 60 minutes on LAPTOP-SMMVETKR, with 2 logon events in a focused 5-minute window during peak attack time, confirms account compromise. Immediate password reset and session termination were required.

5. Attack Timeline Reconstruction

The following timeline was reconstructed by correlating Hydra output, Splunk events, and Windows Event Viewer logs:

Timestamp	Event ID / Tool	Activity
02:02:39 AM	Hydra (Kali)	Brute-force attack initiated against RDP 10.0.2.15:3389 with rockyou.txt
02:02–02:09	4625 (Security)	Multiple failed logon events generated on LAPTOP-SMMVETKR
02:09:21 AM	4624 (Security)	Successful logon recorded — attacker authenticated
02:10:33 AM	4624 (Security)	Second successful logon event observed within 5-minute window
02:12:38 AM	4624 (Security)	Most recent successful logon — active attacker session confirmed
02:14 AM	Splunk UF	Forwarder confirmed running; all events forwarded to SIEM
02:54 AM	4625 / 4624	Continued activity on DESKTOP-HMUS15L: failed + successful logons
02:54:40 AM	4672 (Security)	Special privileges assigned at logon — elevated access confirmed
02:59 AM	Event Viewer	Post-incident cross-validation performed in Windows Event Viewer

6. Windows Event Viewer — Cross-Validation

Windows Event Viewer (Security log) was used to cross-validate the Splunk SIEM findings. At 02:59 AM on 18-03-2026, the following events were confirmed in the Security log on DESKTOP-HMUS15L:

Date/Time	Event ID	Source	Category
18-03-2026 02:54:40	4672	Microsoft Windows security auditing	Special Logon
18-03-2026 02:54:40	4627	Microsoft Windows security auditing	Group Membership
18-03-2026 02:54:40	4624	Microsoft Windows security auditing	Logon
18-03-2026 02:54:39	5379 (x4)	Microsoft Windows security auditing	User Account Management
18-03-2026 02:54:32	4625	Microsoft Windows security auditing	Logon (Failed)
18-03-2026 02:54:30	5379	Microsoft Windows security auditing	User Account Management
18-03-2026 02:54:28	4625	Microsoft Windows security auditing	Logon (Failed)
18-03-2026 02:54:27	5379	Microsoft Windows security auditing	User Account Management

Event 4627 details confirmed: Group membership information was logged with Subject Security ID = SYSTEM, consistent with an elevated privilege logon following successful brute-force authentication. The total Security log count reached 3,642 events, confirming extensive attack activity.

7. Findings & Risk Assessment

7.1 Confirmed Findings

#	Finding	Severity	Evidence
1	RDP brute-force attack successfully launched from Kali Linux using Hydra	CRITICAL	Hydra terminal output
2	Attacker successfully authenticated (Event 4624 confirmed)	CRITICAL	Splunk & Event Viewer
3	121 successful logon events in 60-minute window — anomalous volume	HIGH	Splunk EventCode=4624
4	Elevated privileges granted post-logon (Event 4672)	HIGH	Windows Event Viewer
5	Credential Manager access detected (Event 5379)	HIGH	Windows Event Viewer
6	Attack spanned multiple hosts (LAPTOP-SMMVETKR, DESKTOP-HMUS15L)	MEDIUM	Splunk host field
7	No alerting rule was pre-configured — detection was reactive	MEDIUM	Lab observation

7.2 Recommendations

- Implement Account Lockout Policy: Lock accounts after 5–10 failed attempts within 5 minutes
- Disable RDP on all systems that do not require it; restrict via firewall rules (port 3389)
- Enable Network Level Authentication (NLA) for all RDP connections
- Implement Multi-Factor Authentication (MFA) for all remote access
- Create Splunk alerts for: >5 Event 4625s from same host in 1 minute (brute-force detection)
- Enforce strong password policies: minimum 14 characters, complexity required
- Deploy a honeypot RDP service to detect and divert brute-force attempts
- Regular log review cadence: daily SIEM alert review, weekly anomaly reports

8. Skills Demonstrated

This project demonstrates the following cybersecurity competencies:

Skill Category	Skills Demonstrated
Offensive Security	Hydra brute-force attack, RDP exploitation, password dictionary attacks, Kali Linux tooling
SIEM & Log Analysis	Splunk search queries, event correlation, timeline reconstruction, field extraction and filtering
Windows Security	Windows Security Event IDs (4624, 4625, 4627, 4672, 5379), Event Viewer analysis
Log Management	Splunk Universal Forwarder installation and configuration, WinEventLog source configuration
Incident Response	Attack timeline reconstruction, evidence correlation, cross-tool validation
Documentation	Technical report writing, evidence documentation, risk assessment, portfolio presentation

9. Tools & Technologies

Tool / Technology	Version / Details	Purpose
Kali Linux	Rolling release (VirtualBox VM)	Attacker platform
Hydra (THC-Hydra)	v9.6	RDP brute-force attack tool
rockyou.txt	14,344,399 passwords	Password wordlist for dictionary attack
Windows 10	Version 10.0.26200.6584	Attack target & log source
Splunk Enterprise	Local instance (port 8000)	SIEM for log aggregation and analysis
Splunk Universal Forwarder	Installed on Windows target	Ships Security event logs to Splunk
Windows Event Viewer	Built-in Windows tool	Secondary log validation and cross-check
Oracle VirtualBox	Lab hypervisor	Hosts Kali and Windows VMs

10. Conclusion

This lab project successfully demonstrated the full lifecycle of an RDP brute-force attack — from execution on the attacker machine through to detection, analysis, and incident response using industry-standard tools. The exercise validated the effectiveness of Splunk as a SIEM platform for detecting credential-based attacks through Windows Security event correlation.

The most critical finding — confirmed successful logon events (Event 4624) following extensive failed attempts (Event 4625) — illustrates the real-world danger of weak or default credentials exposed via RDP. The addition of Event 4672 (Special Privileges) confirms the attacker achieved elevated access, which in a production environment would represent a full system compromise.

From a defensive perspective, this project reinforces the value of proactive SIEM alerting, account lockout policies, and MFA as essential controls against brute-force attacks. The skills applied across this project span the full blue-team detection and response workflow, and the offensive exercise provides foundational understanding of attacker methodology essential for effective defence.

— End of Report —

Portfolio Project | RDP Brute-Force Detection Lab | March 2026